# BVS1 – Betriebssysteme

Lecture 1 – Introduction · Winter Semester 2026/2027

Stavros Kousidis

TH Köln · ICCT

# Organization

# How We Work in BVS1

- **Ask questions** and challenge claims.
- **Discuss and collaborate** while learning.
- Treat mistakes and failed attempts as **learning opportunities**.
- Respect your fellow students and their learning.
- Please arrive on time; if you are late, enter quietly.

**Collaboration is encouraged while learning; demonstrated understanding is individual.**

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# Our Learning Goal

By the end of BVS1, you should be able to:

**motivate** fundamental operating-system concepts from the requirements of a computer system,
**explain** how they interact, and
**implement** them in a minimal operating system that provides a simple shell as its user interface.

**motivate** $\longrightarrow$ **explain** $\longrightarrow$ **implement**

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# How BVS1 Works

**Lecture** — develop why

↓

**Exercise** — investigate and reason

↓

**Lab** — work toward the current system milestone

↓

**Checkpoint** — demonstrate and explain

↓

**Electronic exam** — transfer conceptual understanding

You need **basic programming** and a basic machine model.

We refresh **C** as needed and introduce the required **RISC-V assembly** along the way.

**No prior OS-development experience is required.**

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# Weekly Schedule and Support

| | Day | Time | Room |
|---|---|---|---|
| **Lecture** | Friday | 08:50–10:20 | **ZW6-1** |
| **Exercise** | Friday | 10:35–12:20 | **ZW6-1** |
| **Lab** | Wednesday | 08:50–12:20 | **ZO8-6** |
| **Lab** | Friday | 14:05–17:20 | **ZO8-6** |

**The lab takes place every week.**

Only the three checkpoint appointments are mandatory; all other lab sessions are optional, but we are present to help.

Teaching team: Stavros Kousidis · Raphael Ahrens · Anas Jabaly · Abderrahim Benlalam

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# Lab and Mandatory Checkpoints

| Group | Checkpoint 1 | Checkpoint 2 | Checkpoint 3 |
|---|---|---|---|
| M1 | Wed **04 Nov** | Wed **09 Dec** | Wed **20 Jan** |
| M2 | Wed **18 Nov** | Wed **16 Dec** | Wed **13 Jan** |
| F1 | Fri **06 Nov** | Fri **11 Dec** | Fri **15 Jan** |
| F2 | Fri **13 Nov** | Fri **18 Dec** | Fri **22 Jan** |

M = Wednesday · F = Friday

Work individually or in pairs.

In a pair, both students must demonstrate individually that they meet the checkpoint requirements.

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# Checkpoint Rules

- Book checkpoint appointments through **ILU**.
- Changes require a **justified case**; retakes are also organised through ILU.
- A failed checkpoint can be **retaken**.
- A yellow card is recorded for a **failed attempt, late arrival, or no-show**.
- If you expect a delay, email bvs1@f07.th-koeln.de **before the mandatory appointment**.

**Three yellow cards → lab not passed → no exam admission.**

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# Assessment

## Lab

- ungraded,
- three checkpoints must be passed,
- passing the lab is required for **exam admission**.

## Electronic exam

- graded,
- **closed-book, but self-contained**,
- no personal notes, books, scripts or electronic material,
- non-core technical information needed to solve a task is provided with the task.

**The exam tests understanding and transfer, not lookup of terminology.**

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# Help Us Improve BVS1

BVS1 is being developed iteratively.

Your constructive feedback helps us improve the **lecture, exercises and lab** — for you and for future students.

```
your feedback
      ↓
we analyse it
      ↓
we change what should improve
      ↓
we tell you what changed
```

Feedback is welcome throughout the semester; we will also ask for it explicitly at selected points.

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# Operating Systems

# What Are You Operating?

**Particify P1 · Multiple choice**

**Which operating-system environments do you currently use?**

*Select all that apply.*

- Windows
- macOS
- Linux-based systems
- Android
- iOS / iPadOS
- ChromeOS
- Other
- I am not sure



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# Which of These Are Operating Systems?

Particify P2 · Multiple choice



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**Which of these would you currently call an operating system?**

*Select all that apply.*

- Windows
- Linux kernel
- UEFI
- Docker
- Ubuntu
- Android
- QEMU
- Firefox

**Commit first. We will resolve the boundary cases later.**

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# Prediction



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Particify P3 · Prediction

Can a computer execute one program without an operating system?

1. Yes
2. No
3. I am not sure

Remember your answer.

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# Experiment 1 — Hello BVS1

We give a machine one tiny program.

Watch what happens.

```
Hello BVS1
```

**What must have happened before these characters appeared?**

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# What Just Happened? — Think, Then Pair

**Particify P4 · Think**



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Which statements are true?

- QEMU executed our program.
- A virtual CPU executed our program.
- Ubuntu executed our guest program.
- The guest had an operating system.
- QEMU was already running before our guest program started.
- `Hello BVS1` existed somewhere in memory while the program ran.

**Vote individually. Then compare your answers in pairs for two minutes.**

Where do you disagree? What supports your choice?

# Observation

Our program ran.

The machine we just ran had **no operating system of its own**.

Therefore:

> *Executing instructions is not, by itself, what makes an operating system necessary.*

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# What Was Required?

One important idea of the **stored-program / Von Neumann model**:

> *instructions and data are both kept in memory*

```
instructions + data -> memory <-> CPU -> devices
```

Enough to execute and observe a program.

**Still not an operating system.**

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# How Can Software Know What a CPU Understands?

Different processor implementations can run the same machine code only if they agree on a contract.

**ISA — Instruction Set Architecture**

Among other things, an ISA specifies:

- instructions and encodings,
- registers,
- memory-addressing rules,
- exceptional control transfers,
- privilege mechanisms.

For BVS1 we use **RV64: 64-bit RISC-V**.

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# Your Development Machine Is Not Our Target

Your Ubuntu development VM may use **x86-64 or Arm64**.

Our teaching target uses **RV64**.

So there are two different questions:

1. Where do our development tools run?
2. Which machine are we building the program for?

**Development host is not the target machine.**

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# Virtualization vs. Emulation

Can the physical CPU execute the guest / target instructions directly?

## Virtualization

same ISA

```
guest instruction
        ↓
physical CPU
```

Mostly **direct execution**.

Virtualization controls privileged access.

**BVS1:** Multipass → Ubuntu VM

## Emulation

different ISA possible

```
RV64 instruction
        ↓
QEMU
        ↓
host operation
```

QEMU **interprets or translates**.

**BVS1:** QEMU → RV64 target

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# Our Actual Machine Stack

```
physical host
     ↓  virtualization (Multipass)
Ubuntu development VM: bvs1
     ↓  runs
QEMU
     ↓  emulates RV64
RV64 target machine
     ↓  executes
Hello BVS1
```

Only the RV64 target is our teaching machine.

It contains **no guest operating system**.

# The Toolchain Connects Development Host and Target

The **toolchain** is the set of cooperating build tools.

ELF = Executable and Linkable Format

```
source -> compiler / assembler -> object files
       -> linker -> RV64 ELF executable

RV64 ELF -> QEMU -> RV64 target machine
```

**Cross-compilation:** build tools run on the development host and produce code for another target ISA.

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# Change the Problem

Our machine now needs to run **two programs**.

What new problems appear?

Write down as many as you can.

**We will collect them on the board.**

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# Increase the Requirements

Now add:

- many programs,
- multiple users,
- persistent storage,
- keyboard and display,
- network,
- failures,
- programs that do not trust each other.

What responsibilities would you want software to take over?

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# Our Working Definition

An **operating system** is software that helps turn hardware into a usable computer system by providing:

- abstractions,
- resource management,
- coordination,
- protection,
- services for programs and users.

**We will test and sharpen this definition throughout the semester.**

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# What Must We Add to Hello BVS1?

Particify P5 · Open question

What must we add to `Hello BVS1`
before you would call the result an operating system?



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```
Hello BVS1
    ↓
    ?
    ↓
operating system
```

Keep your ideas.

We will return to them during the semester.

# What We Are Going to Build

A small educational operating system on:

RV64 + QEMU `virt`

We start with:

```
Hello BVS1
```

Then add mechanisms **only when a requirement motivates them**.

The long-term target is a **simple shell**.

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# Before You Leave

Challenge these claims:

1. A computer needs an operating system to execute instructions.
2. Virtualization and emulation mean the same thing.
3. An operating system is mainly a graphical user interface.
4. “Operating system” and “processor architecture” describe the same layer.

For each claim:

**agree, disagree, or qualify — and why?**

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# References

Selected references:

- Arpaci-Dusseau & Arpaci-Dusseau, *Operating Systems: Three Easy Pieces*
- Tanenbaum & Bos, *Modern Operating Systems*
- RISC-V ISA specifications
- QEMU RISC-V `virt` documentation
- Canonical Multipass documentation

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# Take-Home Concepts

If you remember five things from today:

- A CPU can execute a program **without a guest operating system**.
- The **ISA** is the machine-level contract between software and processor.
- **Development host** and **target machine** are different roles.
- **Virtualization** and **emulation** solve different problems.
- An **operating system** becomes necessary as requirements for sharing, abstraction, coordination, protection, and services grow.
  *We start with a machine that runs one program — and add OS mechanisms only when a requirement demands them.*

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